CODE SPORT

Sandya Mannarswamy

In this month's column, we feature a set of interview questions on algorithms, data structures, operating systems and computer architecture.

For the past few months, we have been discussing information retrieval, Natural Language Processing (NLP) and the algorithms associated with them. In this month's column, we focus on an important aspect of NLP known as textual entailment.

What is textual entailment?

Let us understand more about this by first looking at the two following snippets of text:

(1) India has a number of nuclear power plants which are used to generate electrical power. The first nuclear power plant was started in Tarapur, Maharashtra. The latest one was commissioned at Kudankulam, Tamilnadu.

(2) Kudankulam in Tamilnadu has a power generation station, which generates electricity.

Now the NLP problem posed is to determine whether Snippet 2 can be inferred from Snippet 1. When human beings parse the text of Snippet 1 and Snippet 2, it is very easy for us to determine whether the latter can be inferred from the former. On the other hand, it is not easy for an automated NLP algorithm to reach this conclusion. This is the problem area that textual entailment techniques attempt to solve. Formally, given a text snippet 'T' and a text snippet representing the hypothesis 'H', a textual entailment program could determine whether they formed a textual entailment pair. 'T' and 'H' form a textual entailment pair, if a human reading 'T' and 'H' would be able to infer 'H' from 'T'.

Consider the following example of two snippets:

(3) The Director of Public Health said, "It is important to stress that this is not a confirmed case of Ebola."

(4) A case of Ebola was confirmed in Mumbai.

Now the question is whether Snippets 3 and 4 constitute a textual entailment pair? The answer of course is obvious to humans—that they do not form a textual entailment pair. But as we will see later, this is not a simple case for an automated textual entailment technique. Many techniques that use surface level string parsing or that are keyword-based, do wrongly mark this pair as textually entailed. Much of the complexity of automatic textual entailment techniques is needed to deal with and avoid such false positives.

An area closely related to textual entailment is paraphrasing. Two statements are paraphrases if they convey almost the same information. Consider the following snippets:

(5) Ayn Rand wrote the book 'Fountainhead'.

(6) Fountainhead was written by Ayn Rand.

(7) Ayn Rand is the author of Fountainhead.

(8) Fountainhead is the work of Ayn Rand.

Now statements 5, 6, 7 and 8 are all paraphrases of each other. If two statements A and B are paraphrases, then they are mutually textually entailed. In other words, both (A, B) and (B, A) are textual entailment pairs. Statement A can be inferred from B and vice versa. But textual entailment does not automatically imply that they are paraphrases. Consider our previous example: Statement 1 implies Statement 2, but not the other way around.

Textual entailment programs can be used for: (a) recognising whether two pairs of natural language expressions form a textual entailment pair, (b) given a single natural language expression, generate all possible TE expressions for it, and (c) given a document or set of documents, extract